YI (Edison) XIE

220 Handan Road, Shanghai, China 200433 • (+86) 18810233420 • yixie22@m.fudan.edu.cn

EDUCATION

Fudan University Shanghai, China
M.S., Academy for Engineering & Technology Expected Jun 2025
Concentrations: Multi-agents System, Reinforcement Learning, Nature Language Processing, Robot Learning.
Advisor: Prof. ZhongXue Gan

Beijing University of Chemical Technology Beijing, China
B.S., Mechanical and Electrical Engineering Sep 2017 - Jun 2021

Beijing University of Chemical Technology Beijing, China
B.S., Economic Management (Dual Degree) Sep 2021 - Jun 2022

PUBLICATION

➢ Towards Efficient and Scalable Multi-Agent Reasoning via Bayesian Nash Equilibrium
Submitted in Oct 2024
- Author: **Yi Xie**, Zhanke Zhou, Chentao Cao, Qiyu Niu, Bo Han.
- Conference: International Conference on Learning Representations (*ICLR 2025*)

➢ Refined Sequential Execution for Diverse Multi-Agent Systems in Cooperative Tasks.
Resubmitted in Oct 2024 (IJCAI 7764 rejected in Feb 2024)
- Author: **Yi Xie**, Ziqing Zhou, Chun Ouyang, Siao Liu, Zhile Zhao, Zhongxue Gan
- Conference: the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)

➢ Heuristics-Assisted Experience Replay Strategy for Cooperative Multi-Agent Reinforcement Learning
Submitted in Oct 2024
- Author: **Yi Xie**, Ziqing Zhou, Chun Ouyang, Siao Liu, Zhile Zhao, Zhongxue Gan
- Conference: the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)

➢ Learning from Past with Multi-view Attention for Robotic Manipulation.
Submitted in Mar 2024
- Author: **Yi Xie**, Gaoxiong Lu, Zhongxue Gan
- Confer*ence: 2025 IEEE International Conference on Robotics & Automation (ICRA 2025*)

➢ A Hierarchical Framework for Multi-Agent Navigation in Complex Environments.
Submitted in Nov 2023
- Author: **Yi Xie**, Ziqing Zhou, Chun Ouyang, Siao Liu, Zhile Zhao, Bo Chen, Zhongxue Gan
- Journal: IEEE Transactions on Neural Networks and Learning Systems (*TNNLS*)

➢ A Framework for Dynamical Distributed Flocking Control in Dense Environments.
Published in Nov 2023
- Author: Ziqing Zhou, Chun Ouyang, Linqiang Hu, **Yi Xie**, Yuning Chen, Zhongxue Gan
- Journal: Expert Systems with Applications

➢ Improving Generalization in Visual Reinforcement Learning via Conflict-aware Gradient Agreement Augmentation.
Published in Jul 2023
- Author: Siao Liu, Zhaoyu Chen, Yang Liu, Yuzheng Wang, Dingkang Yang, Zhile Zhao, Ziqing Zhou, **Yi Xie**, Wei Li, Wenqiang Zhang, Zhongxue Gan
- Conference: IEEE/CVF International Conference on Computer Vision (*ICCV*), 2023

RESEARCH EXPERIENCE

Reinforcement Learning Based Post Training and Searching toward Enhancing Reasoning Ability for General LLM Hongkong

Hong Kong Baptist University, TMLR Group Mar 2024 - Further

- Conceptually formalize BNE in multi-LLM systems and technically instantiate it through a hierarchical optimization framework EcoNash to improve reasoning over collaboration of LLMs
- Address the non-trivial challenge of scaling up multi-LLM systems with local-global Nash, facilitated by low reliance on inter-agent communication and convergence guarantee.
- Further exploration of RL post-training and searching to boost the reasoning capabilities of language models aims to achieve lower consumption and higher reliability in general LLMs

Multi-Agents Coordination for Scalable Swarm via Reinforcement Learning, Graph Neural Networks and Heuristic Algorithm

Shanghai, China

Fudan university, Academy for Engineering & Technology Jan 2022 – Dec 2024

- Developed multi-agent coordination strategies leveraging reinforcement learning to enhance swarm scalability and efficiency, aiming to improve collective performance in dynamic environments.
- Utilized acyclic graph-based graph neural networks to effectively solve multi-agent coordination problems, enabling structured communication and robust decision-making among agents.
- Implemented CMOSO heuristic algorithms to address credit assignment problem in multi-agent reinforcement learning, ensuring fair reward distribution to facilitate cooperative behavior.

Precision Robotic Movement through Imitation Learning and VRL: Keyframe Analysis, Multi-Modal Inputs, and Spatiotemporal Attention Guangzhou, China

Jihua lab, Robotics lab Jun 2022 - Oct 2022

- Conducted research on achieving high-precision robotic arm movements by leveraging imitation learning and visual reinforcement learning (VRL) techniques to replicate expert demonstrations.
- Utilized keypoint analysis and genetic algorithms to extract crucial keyframes from demonstrations, integrating multi-modal inputs, along with semantic details, to enrich the learning dataset.
- Developed a decision-making network based on spatiotemporal attention mechanisms to process integrated feature and accurately predict next-step actions, ensuring precise robotic arm movements.

HONORS & AWARDS

Individual Scholarship ×7 (4 for Second prize, 3 for others) 2018-2024
"Challenge Cup" National College Student Business Plan Competition (Top 0.5% in China) 2022
ACM-ICPC(ACM International Collegiate Programming Contest) Asia Beijing Silver 2022

SKILLS

Technical Skills: Python torch ROS(basic)
Language: Mandarin(native) English(TOEFL93(next test at Oct)) France(basic)

Services

Reviewer for Neurips 2024/ ICLR 2025/